
EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 3A

1. Signal Fundamental Frequency

For each of the discrete-time signals below, determine the fundamental period (and corresponding fundamental frequency), or explain why the signal is not periodic.

(a) $x[n] = e^{i\frac{\pi}{5}n}$ for all integers n .

(b) $x[n] = \cos\left(\frac{3\pi}{5}n\right)$ for all integers n .

(c) $x[n] = \sin(n)$ for all integers n .

2. Orthogonal Building Blocks

For an integer $k \in \{0, 1, 2\}$, define the vector $\phi_k \in \mathbb{C}^3$ by

$$\phi_k[n] = e^{-ik\frac{2\pi}{3}n}, \quad n = 0, 1, 2.$$

Recall the inner product of two such vectors is $\langle a, b \rangle = \sum_{n=0}^2 a[n] b^*[n]$.

- (a) Write out the 3 entries of ϕ_0, ϕ_1, ϕ_2 , then compute $\langle \phi_0, \phi_1 \rangle$ and $\langle \phi_1, \phi_2 \rangle$. What do you notice?

Hint: Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, will help you evaluate each entry.

- (b) Explain why ϕ_0, ϕ_1, ϕ_2 can be used to construct *any* vector in $\mathbb{C}^3$ as a linear combination of them.